NADINE PIGIDA

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EXPERIENCE

COMPUTATIONAL BIOLOGY RESEARCH STUDENT
SICKKIDS HOSPITAL • SEP 2023 – AUG 2024

- Used explainable AI methods (SHAP) to uncover significant genes in an ensemble convolutional neural network trained to classify pediatric cancers
- Trained XGBoost models on RNA-seq data to extract gene-level explanations
- Explored the use of CycleGANs in generating normal samples to pair with tumor samples for input into bioinformatics tools
- Worked on a high-performance computing cluster and utilized Slurm effectively for testing and running tools in whole genome analysis pipeline
- Optimized the structural variant subworkflow of the pipeline using WDL, and implemented a 'tumor sample only' mode, reducing runtime by ~24h

TEACHER'S ASSISTANT - MACHINE LEARNING (EDAN96)
LUND UNIVERSITY • OCT – DEC 2025

- Guided students through core ML concepts (K-Means, KNN, decision trees, logistic regression, neural networks, CNNs) and foundational theory (probability, entropy, information theory).
- Graded weekly lab assignments orally and provided feedback.

SOFTWARE ENGINEER
DOCTOR INBOX • MAY – AUG 2022, JAN – APR 2023

- As a founding employee, developed an application that automates doctors' time-consuming administrative work, cutting prescription refill time by over 75%
- Used AWS to create and train a custom ML model for extracting relevant data from scanned prescriptions
- Transformed the local application into a production-level system using AWS (Lambda and DocumentDB)
- Designed and developed a second application to automate billing processes for family doctors

SOFTWARE DEVELOPER
CULTUREALLY • SEP 2021 – DEC 2021

- Investigated and resolved bugs impacting various frontend and backend tasks
- Wrote SQL queries to report live per-account engagement metrics to admin users
- Designed new UI/UX for delivering employee training modules through the web app

SKILLS

- **Languages:** Python, Bash, WDL, SQL, C/C++, Kotlin, HTML, CSS, JavaScript
- **Technologies:** AWS (Lambda, Sagemaker, Comprehend, S3, DocumentDB, DynamoDB), Git/GitHub, HPC/Slurm, Google Firebase, Linux, Docker

CONFERENCE PRESENTATIONS

- Kharseh, B., Pigida, N. et al.
  Co-essentiality patterns reveal novel gene targets in childhood cancer.
  Temerty Faculty of Medicine Research Showcase, 2024. *(Oral presentation)*

PROJECTS

CAPUCHIN BIRD CALL DETECTION

- Built a convolutional neural network in Python/TensorFlow to detect and count Capuchin bird calls from audio

KINDKART

- Chrome extension built with Google Firebase to help users make sustainable purchases online
- 1st place in Industry category, SEThacks hackathon

EDUCATION

Bachelor of Computer Science 88% GPA
UNIVERSITY OF WATERLOO • APRIL 2025

Master of Science in Bioinformatics
LUND UNIVERSITY • EXPECTED APRIL 2027